

Stock Market Fundamentals Index

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Abstract

We introduce the Market Fundamentals Index (MFI), a novel metric constructed from a tensor representation of firm-level accounting data and smoothed via Tucker decomposition. The MFI captures time-varying dispersion in market fundamentals, and we demonstrate its strong statistical dependence on a price-based financial chaos index (FCIX). Building on this tensor framework, we develop a predictive pipeline that imputes missing fundamentals through mask-aware Tucker decomposition and forecasts the full next-quarter panel via low-rank CANDECOMP/PARAFAC regression deployed as a booster over fixed-effects and per-feature ridge baselines. On a calendar-fixed out-of-time holdout the tensor improves on a fully tuned ridge benchmark, its gains concentrating on non-stationary line items, and the improvement survives a locked-hyperparameter transfer to the full 499-firm selection universe. A sequence of hypotheses specified ex ante then traces where markets price this information: in mega-cap investment-grade names, announcement returns, option-implied volatility, and CDS spreads all subsume the tensor’s signals, while in a dedicated high-yield/crossover universe, persistent cash-flow over-performance against the tensor forecast predicts economically meaningful CDS spread tightening (approximately 6.5 basis points per standard deviation per quarter) in all four model configurations—evidence consistent with slower incorporation of this information where default risk is material.

Keywords: multilinear tensor regression, Tucker decomposition, CANDECOMP/PARAFAC, market fundamentals, volatility index, credit default swaps, event study, market efficiency

1 Introduction

This paper investigates asset price fluctuations and their relationship with underlying firm fundamentals. Incorporating fundamental accounting data is essential for capturing the full scope of publicly available market information. Together with news-based uncertainty measures and historical prices, these three information sources form the basis for rigorous tests of the efficient market hypothesis across multiple time horizons.

The practical significance is direct: if the market exhibits inefficiency at daily, monthly, or quarterly frequencies, investors can leverage historical data to forecast future asset trajectories. Conversely, if the market proves efficient, prediction-based models must give way to alternative asset allocation strategies. Our framework therefore provides actionable guidance for traders and portfolio managers seeking to position investments optimally.

1.1 Notations and Preliminaries

Throughout this paper, scalar quantities, vectors, and matrices are denoted by lowercase lightface (e.g., x), lowercase boldface (e.g., $\mathbf{x}$), and uppercase boldface (e.g., $\mathbf{X}$) letters, respectively. All vectors are assumed to be column vectors unless stated otherwise. Boldface calligraphic letters (e.g., $\mathcal{X}$) are reserved for tensors, while their lightface calligraphic counterparts (e.g., $\mathcal{X}$) typically denote induced sample spaces.

Tensors are algebraic structures that generalize matrices to higher dimensions. The number of dimensions of a tensor is referred to as its order or mode. For instance, $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ denotes an N th order tensor with $K = \prod_{n=1}^N I_n$ elements in total. Analogous to matrix rows and columns, the mode- n fibers of a tensor are obtained by fixing all indices except the n th one. Fixing all but two indices yields the slices of the tensor.

The Frobenius norm of a tensor $\mathcal{X}$ is defined by

$$\|\mathcal{X}\|_F = \sqrt{\langle \mathcal{X}, \mathcal{X} \rangle} = \sqrt{\sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2} \dots \sum_{i_N=1}^{I_N} x_{i_1 i_2 \dots i_N}^2}. \quad (1)$$

To reduce tensor order and facilitate matrix computations, one often resorts to unfolding (also called matricization). The vector unfolding of $\mathcal{X}$, denoted by

$$\mathbf{x} = \text{vec}(\mathcal{X}) \in \mathbb{R}^K, \quad (2)$$

reshapes the tensor into a vector. The mode- n matricization of $\mathcal{X}$, given by

$$\mathcal{X}_{(n)} = [\mathbf{f}_1^{(n)}, \mathbf{f}_2^{(n)}, \dots, \mathbf{f}_{K/I_n}^{(n)}], \quad (3)$$

stacks the mode- n fibers as columns of a matrix of size $I_n \times (K/I_n)$.

The mode- n product of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ with a matrix $\mathbf{U} \in \mathbb{R}^{J \times I_n}$ is denoted by $\mathcal{X} \times_n \mathbf{U}$ and defined element-wise as

$$(\mathcal{X} \times_n \mathbf{U})_{i_1 \dots i_{n-1} j i_{n+1} \dots i_N} = \sum_{i_n=1}^{I_n} x_{i_1 \dots i_N} u_{j i_n}, \quad (4)$$

yielding a tensor of dimensions $I_1 \times \dots \times I_{n-1} \times J \times I_{n+1} \times \dots \times I_N$. This operation is equivalent to multiplying $\mathbf{U}$ from the left on each mode- n fiber as follows:

$$\mathcal{Y} = \mathcal{X} \times_n \mathbf{U} \iff \mathcal{Y}_{(n)} = \mathbf{U} \mathcal{X}_{(n)}. \quad (5)$$

When dealing with third-order tensors $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, a useful perspective is to treat them as stacks of frontal slices $\mathcal{X}^{(n)} \in \mathbb{R}^{I_1 \times I_2}$ for $n = 1, 2, \dots, I_3$. Define the operator

$$\text{unfold}(\mathcal{X}) := [\mathcal{X}^{(1)}, \mathcal{X}^{(2)}, \dots, \mathcal{X}^{(I_3)}]^\top, \quad (6)$$

which stacks all frontal slices vertically. The corresponding inverse operator **fold** satisfies

$$\text{fold}(\text{unfold}(\mathcal{X})) := \mathcal{X}. \quad (7)$$

Two widely used tensor decompositions are the Tucker and the polyadic (or CP) decompositions. The Tucker decomposition approximates a third-order tensor $\mathcal{X}$ by a dense core tensor $\mathcal{H} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ and factor matrices $\mathbf{A} \in \mathbb{R}^{I_1 \times R_1}$, $\mathbf{B} \in \mathbb{R}^{I_2 \times R_2}$, and $\mathbf{C} \in \mathbb{R}^{I_3 \times R_3}$ as follows:

$$\mathcal{X} \approx \hat{\mathcal{X}} = \mathcal{H} \times_1 \mathbf{A} \times_2 \mathbf{B} \times_3 \mathbf{C}. \quad (8)$$

Alternatively, the polyadic decomposition expresses the tensor as a sum of rank-one components,

$$\mathcal{X} \approx \hat{\mathcal{X}} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r, \quad (9)$$

where the symbol $\circ$ denotes the vector outer product and R is the rank of the approximation. When R is the minimal number of components required to achieve equality, the decomposition is referred to as the canonical polyadic decomposition (CPD).

Finally, a third-order tensor $\mathcal{X}$ is said to be rank-one if it admits the representation

$$\mathcal{X} = \mathbf{a} \circ \mathbf{b} \circ \mathbf{c}, \quad (10)$$

for some vectors $\mathbf{a} \in \mathbb{R}^{I_1}$, $\mathbf{b} \in \mathbb{R}^{I_2}$, and $\mathbf{c} \in \mathbb{R}^{I_3}$.

2 Encoding Fundamentals Information

In this section, we develop a methodology for embedding stock market fundamentals into structured representations and identifying a subset of features that preserves the latent economic structure of the market. The ultimate objective is to facilitate the construction of a volatility index grounded in fundamental information. The selected

features are required to align with a reference taxonomy of economic activity, namely the Global Industry Classification Standard (GICS), thus providing a bridge between qualitative categorization and quantitative representation.

2.1 Feature Selection via Clustering

Let $S = \{S_1, S_2, \dots, S_N\}$ denote the collection of assets observed in the market, and let $F = \{f_1, f_2, \dots\}$ denote the universe of available fundamental features. For each asset S_i , let $\mathbf{d}_i \in \mathbb{R}^{|F'|}$ denote the vector of observed values for a subset $F' \subset F$. These vectors are assembled into a design matrix $\mathbf{D} \in \mathbb{R}^{N \times |F'|}$ defined as

$$\mathbf{D} = \begin{bmatrix} \mathbf{d}_1^\top \\ \mathbf{d}_2^\top \\ \vdots \\ \mathbf{d}_N^\top \end{bmatrix} = \begin{matrix} & \begin{matrix} f_1^* & f_2^* & \dots & f_{|F'|}^* \end{matrix} \\ \begin{matrix} S_1 \\ S_2 \\ \vdots \\ S_N \end{matrix} & \begin{bmatrix} d_{11} & d_{12} & \dots & d_{1|F'|} \\ d_{21} & d_{22} & \dots & d_{2|F'|} \\ \vdots & \vdots & \ddots & \vdots \\ d_{N1} & d_{N2} & \dots & d_{N|F'|} \end{bmatrix} \end{matrix}.$$

Our objective is to identify a subset $F^* \subset F$ whose induced clustering structure naturally mirrors the established GICS sub-industry classification. Formally, let $\mathcal{C}(F')$ denote the number of clusters inferred from $\mathbf{D}$ via a clustering algorithm, and let $\mathcal{G}$ denote the cardinality of the GICS sub-industry partition (i.e., $\mathcal{G} = 158$). The optimal subset is then

$$F^* = \arg \min_{F' \subset F} |\mathcal{C}(F') - \mathcal{G}|. \quad (11)$$

This defines our best-match criterion, which balances parsimony in feature selection with fidelity to external economic structure. A common alternative to this discrete-matching approach is to measure partition similarity using adjusted Rand index or normalized mutual information, though such refinements are omitted here in favor of interpretability.

Traditional clustering algorithms such as k -means require the number of clusters to be specified a priori, which is incompatible with our goal of estimating structure from the data itself. To address this, we employ the affinity propagation algorithm [1], which treats every observation as a potential cluster exemplar and seeks to minimize an energy function through iterative message passing. This framework automatically determines the number of clusters, with control exerted through a tunable preference parameter. Technical details and convergence analysis are provided in [2].

An additional challenge lies in the representation of temporal information. The matrix $\mathbf{D}$ discards the temporal dynamics of each fundamental feature by aggregating values into a single statistic per asset. While this introduces a form of statistical coarsening, it is justified by the relative stationarity of taxonomic classification over short horizons. Sector and sub-industry affiliations typically undergo redefinition on a 4-5 year timescale, often in response to shifts in macroeconomic conditions or technological innovation. To reflect this, we construct $\mathbf{D}$ by averaging quarterly values of each feature from the first quarter of 2016 to the fourth quarter of 2019.¹

¹Fundamental data retrieved from Compustat via the WRDS platform.

For each candidate subset $F' \subset F$, the design matrix $\mathbf{D}$ is constructed and subjected to affinity propagation clustering. The resulting number of clusters is then compared to $\mathcal{G} = 158$. The subset F^* is selected according to the best-match criterion. While dimensionality reduction methods such as PCA, Lasso, or autoencoders could be considered, we intentionally avoid these to preserve economic interpretability of the features. Moreover, robustness checks revealed that F^* remained stable across perturbed data subsets, indicating selection consistency.

In total, over 600 distinct quarterly and annual features were considered. Through extensive search and evaluation, a 40-feature subset F^* was identified, as detailed in Table 2. The affinity propagation algorithm applied to $\mathbf{D}$ constructed from F^* yielded 156 clusters, just two fewer than the GICS-defined sub-industry count. This near match supports the claim that F^* provides a structurally faithful representation of economic heterogeneity across the market. As a supplementary check, Table 1 reports direct partition-similarity metrics against GICS levels on the canonical affinity-propagation design matrix; the agreement is positive but imperfect, so we interpret the feature set as structurally aligned rather than as a reconstruction of the taxonomy.

Table 1: Supplementary alignment metrics between affinity-propagation clusters and GICS labels. ARI denotes adjusted Rand index, NMI normalized mutual information, AMI adjusted mutual information, and purity the share of firms assigned to the modal GICS class within each cluster.

GICS level	Classes	AP clusters	ARI	NMI	Purity
Sector	11	27	0.135	0.349	0.474
Group	23	27	0.182	0.438	0.388
Industry	55	27	0.140	0.531	0.294
Sub-industry	94	27	0.121	0.578	0.260

2.2 Market Fundamentals Index

We now introduce a tensor-based volatility index that summarizes the aggregate dynamics of stock market fundamentals over time. This Market Fundamentals Index (MFI) parallels the FCIX developed in prior literature [3], but it is uniquely derived from fundamental accounting data rather than asset returns.

Let $\mathcal{D} \in \mathbb{R}^{N \times |F^*| \times |\mathcal{T}|}$ denote a third-order design tensor storing the temporal trajectories of selected fundamental features for N assets, where $|\mathcal{T}|$ is the number of time periods. In the empirical construction below, $\mathcal{D}$ contains $N = 499$ firms, $|F^*| = 40$ quarterly accounting features, and $|\mathcal{T}| = 140$ quarters from 1990Q1 through 2024Q4, with an observed-entry density of 74.0%. The structure of this tensor is illustrated in Figure 1.

Table 2: Selected fundamental features used for constructing the design matrix.

Feature	Feature
Acquisitions	Inventories - Total
Assets - Other - Total	Investing Activities - Net Cash Flow
Assets and Liabilities	Investing Activities - Other
Capital Expenditures	Long-Term Debt - Total
Cash and Short-Term Investments	Long-Term Debt - Issuance
Cash Dividends	Liabilities Netting Other Adjustments
Common/Ordinary Equity - Total	Noncontrolling Interest
Comprehensive Income - Total	Non-Operating Income (Expense) - Total
Cost of Goods Sold	Operating Activities - Net Cash Flow
Debt in Current Liabilities	Operating Income Before Depreciation
Earnings Per Share (Basic)	Preferred/Preference Stock (Capital) - Total
Earnings Per Share (Diluted)	Pretax Income
Excess Tax Benefit of Stock Options	Receivables - Total
Extraordinary Items	Sale of Common and Preferred Stock
Financing Activities - Net Cash Flow	Sale of Investments
Funds from Operations - Other	Sale of PPE and Investments - Gain/Loss
Quarterly Income Before Extraordinary Items	Sales/Turnover (Net)
Annual Income Before Extraordinary Items	Short-Term Investments - Total
Income Taxes	Special Items
Intangible Assets - Total	Stockholders Equity

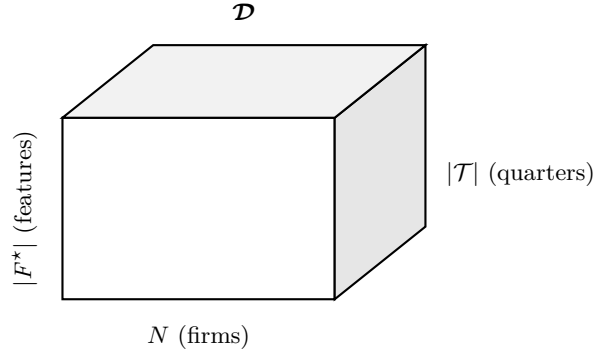


Fig. 1: Schematic of a third-order market fundamentals design tensor.

To characterize the structural complexity of $\mathcal{D}$ and inform the construction of MFI, we begin by applying a polyadic decomposition as follows:

$$\mathcal{D} \approx \hat{\mathcal{D}} = \sum_{r=1}^R \mathbf{s}_r \circ \mathbf{f}_r \circ \mathbf{t}_r, \quad (12)$$

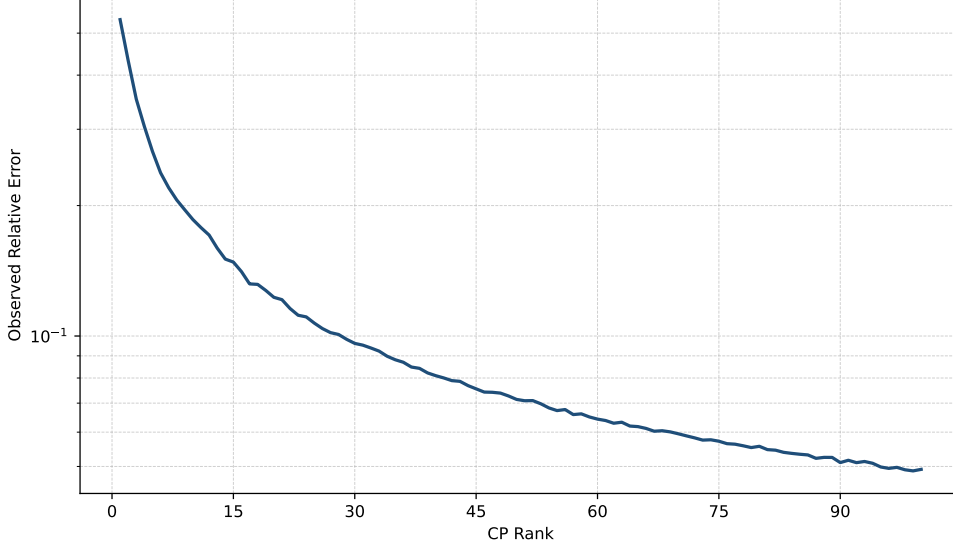


Fig. 2: Plot of the relative error of polyadic decomposition of $\mathcal{D}$ for different values of R .

where $\mathbf{s}_r \in \mathbb{R}^N$, $\mathbf{f}_r \in \mathbb{R}^{|F^*|}$, and $\mathbf{t}_r \in \mathbb{R}^{|\mathcal{T}|}$ denote the r th rank-one component. As shown in Figure 2, the relative error of approximation decays slowly in R : on the order of one hundred rank-one terms are required before the observed-entry error approaches the 5% level. Thus $\mathcal{D}$ is not well approximated by a small number of additive rank-one components, suggesting that the interactions among stocks, fundamentals, and time exhibit non-separable and highly multiway structure.

Motivated by this observation, we instead employ Tucker decomposition, which generalizes matrix SVD to higher-order tensors by introducing a core tensor to capture multiway interactions. Specifically, we write

$$\mathcal{D} \approx \hat{\mathcal{D}} = \mathcal{H} \times_1 \mathbf{S} \times_2 \mathbf{F} \times_3 \mathbf{T}, \quad (13)$$

where $\mathcal{H} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ is the core tensor, and $\mathbf{S} \in \mathbb{R}^{N \times R_1}$, $\mathbf{F} \in \mathbb{R}^{|F^*| \times R_2}$, and $\mathbf{T} \in \mathbb{R}^{|\mathcal{T}| \times R_3}$ are the factor matrices. The multilinear rank vector is defined as $\mathbf{R} = [R_1, R_2, R_3]^\top$. The mask-aware Tucker decomposition is obtained by solving the observed-entry optimization problem

$$\min_{\mathcal{H}, \mathbf{S}, \mathbf{F}, \mathbf{T}} \|\mathbf{M} \odot (\mathcal{D} - \mathcal{H} \times_1 \mathbf{S} \times_2 \mathbf{F} \times_3 \mathbf{T})\|_F, \quad (14)$$

where $\mathbf{M}$ is the binary observation mask and $\odot$ denotes elementwise multiplication.

To preserve economic interpretability, we set R_1 equal to the number of observed GICS industry codes in the expanded sample and retain the full feature dimension in the second mode. This yields $R_1 = 67$ and $R_2 = 40$. We then fix $R_3 = 20$ to

summarize the dominant temporal modes and evaluate the observed-entry relative approximation error defined by

$$\text{rel}_{\text{error}} = \frac{\|\mathbf{M} \odot (\mathcal{D} - \hat{\mathcal{D}})\|_F}{\|\mathbf{M} \odot \mathcal{D}\|_F}. \quad (15)$$

Rather than minimizing this error to the lowest achievable level, we intentionally select a moderate value of R_3 that filters out noise and captures only the most persistent components. Based on our empirical evaluation, we use the multilinear rank vector $[67, 40, 20]$, which yields an observed-entry relative approximation error of 5.45%.

Supplementary stability checks support the construction choice. Using the saved v_2 time-factor matrix, replacing the mean-absolute aggregation in Definition 1 with an RMS aggregation gives a highly correlated index ($\rho = 0.968$); a single-factor version is less stable ($\rho = 0.581$), which supports aggregating across all temporal factors rather than selecting one component. Nearby rank/normalization checks remain close to the baseline MFI (minimum correlation 0.938, with a no-RMS $[67, 20, 20]$ variant at $\rho = 0.996$). An initialization audit at rank $[67, 20, 20]$ shows that SVD initialization reaches an observed-entry error near 5.5% across iteration budgets, whereas random starts remain far above this error under the same budgets; accordingly, the reported MFI uses the SVD-initialized Tucker solution.

The temporal factor matrix $\mathbf{T}$ now contains R_3 latent time series of length $|\mathcal{T}|$. Each column captures a distinct latent mode of variation in the temporal dimension of the tensor. To aggregate these into a single index, we define the Market Fundamentals Index by taking the average magnitude across the R_3 components at each time t , whose formal definition is provided below.

Definition 1 (MFI). Let $\mathcal{D}$ be a third-order tensor of market fundamentals smoothed via Tucker decomposition as $\hat{\mathcal{D}} = \mathcal{H} \times_1 \mathbf{S} \times_2 \mathbf{F} \times_3 \mathbf{T}$, with $\mathbf{T} \in \mathbb{R}^{|\mathcal{T}| \times R_3}$. The Market Fundamentals Index (MFI) is defined by

$$\text{MFI}(t) := \frac{1}{R_3} \sum_{k=1}^{R_3} |\mathbf{T}_{tk}|, \quad (16)$$

for $t = 1, \dots, |\mathcal{T}|$.

The absolute value ensures that phase shifts or sign reversals do not cancel signal strength, and the averaging operation yields a smoothed, aggregate representation of the total volatility embedded in the fundamentals. As in the case of FCIX_t , we denote the time series realization of this statistic by MFI_t .

Figure 3 displays the quarterly realizations of MFI from January 1990 through December 2024. The time series reveals distinct structural episodes, marked by substantial shifts in the volatility of firm-level fundamentals. The index rises gradually through the 1990s from a low and stable base, with a first pronounced local peak around the 2001 recession and dot-com unwind. The most elevated readings of the

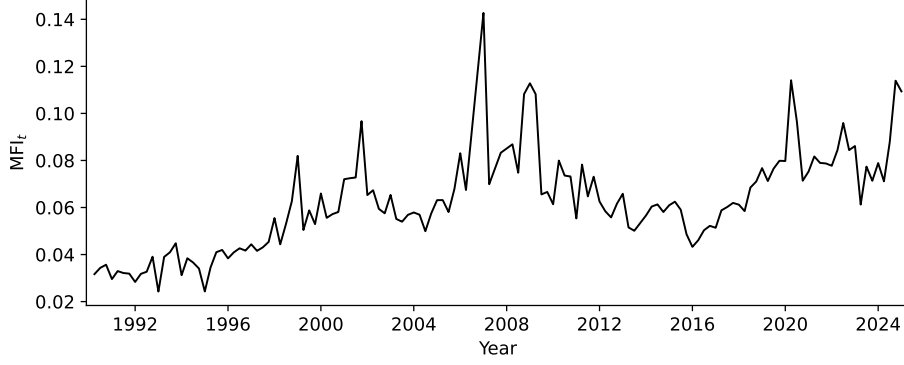


Fig. 3: Plot of the quarterly MFI over the period January 1990–December 2024.

sample occur in 2006–2009: the index spikes sharply in late 2006 and remains elevated throughout the 2007–2009 financial crisis, reflecting heightened dispersion and instability in accounting fundamentals consistent with widespread balance sheet stress across sectors.

Post-crisis, the MFI declines steadily, reaching its calmest levels around 2015–2016, where it enters a low-volatility regime with relatively minor fluctuations. An upward drift is again visible beginning in 2017–2018, culminating in a sharp spike in the first quarter of 2020 at the onset of the pandemic, and the index climbs once more toward the end of the sample in 2024. These patterns suggest that MFI effectively captures systemic dislocations in fundamental data and exhibits temporal alignment with major financial crises. The dynamics of this index and its potential predictive content are further explored in subsequent empirical tests.

3 Statistical Dependence Between MFI and FCIX

We now examine the empirical relationship between the MFI and the FCIX, with the goal of determining whether their co-movement reflects genuine statistical dependence. Identifying such dependence would support the hypothesis that fluctuations in market fundamentals are structurally linked to volatility in asset prices, thereby providing empirical grounding for the efficient market hypothesis explored in the following sections.

We begin by computing the sample cross-correlation between the quarterly time series $\{MFI_t\}$ and $\{FCIX_t\}$ over a range of lags. Figure 4 illustrates the resulting correlation coefficients from -10 to $+10$ quarters. The strongest association is contemporaneous, with a peak correlation of approximately 0.32 at lag zero, indicating a moderate near-term alignment between price-based volatility and variation in accounting fundamentals.

While correlation provides a preliminary indication of synchrony, it is limited to capturing linear dependence and cannot rule out the possibility of spurious alignment. To test for general dependence, including nonlinear and higher-order structures,

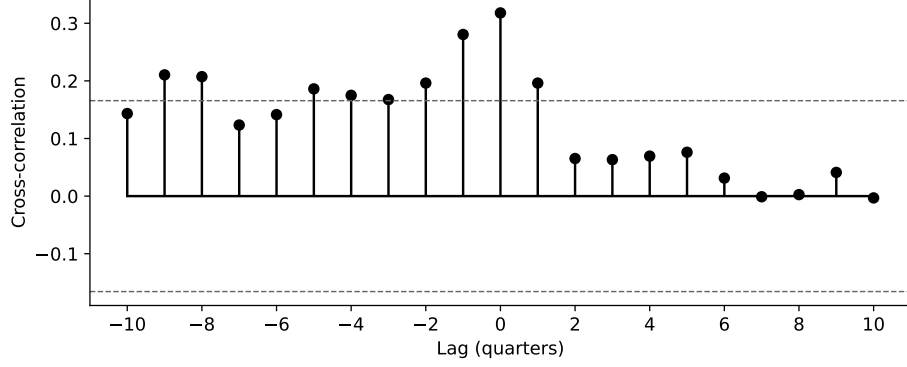


Fig. 4: Sample cross-correlation between quarterly FCIX_t and MFI_t over the period January 1990–December 2024.

we employ a nonparametric framework based on empirical distribution comparisons. Specifically, we adopt the consistent testing methodology proposed in [4, 5], which compares the joint empirical distribution with the product of its marginals using two divergence-based test statistics.

Let $\{(\text{FCIX}_t, \text{MFI}_t)\}_{t=1}^n$ be a sequence of i.i.d. observations drawn from an unknown joint distribution $\mathbb{G}_{\text{FCIX}, \text{MFI}}$. Denote the corresponding marginal distributions by $\mathbb{G}_{\text{FCIX}}$ and $\mathbb{G}_{\text{MFI}}$. The null hypothesis of independence is expressed as:

$$H_0^{\text{INDP}} : \mathbb{G}_{\text{FCIX}, \text{MFI}} = \mathbb{G}_{\text{FCIX}} \times \mathbb{G}_{\text{MFI}}.$$

Let $\mathcal{P}_n = \{A_{n,1}, \dots, A_{n,m_n}\}$ and $\mathcal{Q}_n = \{B_{n,1}, \dots, B_{n,m'_n}\}$ denote measurable partitions of $\mathbb{R}_{\geq 0}$, constructed to discretize the support of the empirical distributions. The first test statistic is based on the L_1 distance between the joint and product measures:

$$L_n := \sum_{A \in \mathcal{P}_n} \sum_{B \in \mathcal{Q}_n} |\mathbb{G}_{n, \text{FCIX}, \text{MFI}}(A \times B) - \mathbb{G}_{n, \text{FCIX}}(A) \mathbb{G}_{n, \text{MFI}}(B)|. \quad (17)$$

The empirical probabilities are defined by:

$$\mathbb{G}_{n, \text{FCIX}, \text{MFI}}(A \times B) := \frac{1}{n} \sum_{t=1}^n \mathbb{I}[(\text{FCIX}_t, \text{MFI}_t) \in A \times B], \quad (18)$$

$$\mathbb{G}_{n, \text{FCIX}}(A) := \frac{1}{n} \sum_{t=1}^n \mathbb{I}[\text{FCIX}_t \in A], \quad (19)$$

$$\mathbb{G}_{n, \text{MFI}}(B) := \frac{1}{n} \sum_{t=1}^n \mathbb{I}[\text{MFI}_t \in B]. \quad (20)$$

As an alternative, we compute a log-likelihood divergence statistic, which measures the Kullback–Leibler distance between the empirical joint and product-marginal

distributions:

$$I_n := \sum_{A \in \mathcal{P}_n} \sum_{B \in \mathcal{Q}_n} \mathbb{G}_{n, \text{FCIX}, \text{MFI}}(A \times B) \log \left(\frac{\mathbb{G}_{n, \text{FCIX}, \text{MFI}}(A \times B)}{\mathbb{G}_{n, \text{FCIX}}(A) \mathbb{G}_{n, \text{MFI}}(B)} \right). \quad (21)$$

Under the null hypothesis of independence, both L_n and I_n converge in distribution, and critical values can be approximated either via permutation or using asymptotic theory.

Table 3 reports the results of both tests applied to $\{\text{FCIX}_t\}_{t=1}^{140}$ and $\{\text{MFI}_t\}_{t=1}^{140}$, using $m_n = m'_n = 6$ equal-frequency discretization bins per dimension. Critical thresholds are obtained from 10,000 random permutations of the paired quarterly series. As a serial-dependence check, we also repeat the null simulation using circular block permutations of the MFI series with four- and eight-quarter block lengths.

Table 3: Results of independence tests between quarterly FCIX_t and MFI_t .

	L_1 -based	Log-likelihood
Test statistic	0.546	0.237
Critical thresholds		
1% level	0.459	0.174
5% level	0.421	0.148
10% level	0.403	0.136
Block-permutation p -values		
4-quarter blocks	0.003	0.006
8-quarter blocks	0.025	0.043

Both test statistics exceed their respective iid-permutation critical values at the 1% significance level, leading to the rejection of the null hypothesis H_0^{INDP} . The block-permutation check weakens the rejection, as expected for persistent quarterly series, but still rejects at conventional levels (1% for four-quarter blocks, 5% for eight-quarter blocks). This provides evidence that FCIX_t and MFI_t are dependent, indicating that price-based market volatility and structural shifts in accounting fundamentals are jointly distributed and possibly mutually informative. These findings serve as a critical empirical foundation for evaluating market efficiency in the subsequent section.

4 Predictive Modeling for Fundamentals

Having established the MFI as a summary of cross-sectional dispersion in fundamentals and demonstrated its statistical dependence with FCIX, we now assess whether the underlying tensor structure contains predictive information for next-quarter accounting values. The task is to forecast a firm-by-feature target matrix using the preceding L quarters of fundamentals, with $L \in \{2, 4\}$ corresponding to 6-month and 1-year lookbacks. Methodologically, this places us in the tensor-on-tensor regression literature [6–8]: rather than fitting a separate time-series model per accounting

line item, a single low-rank multilinear map is estimated jointly across firms, features, and lags, so that predictive strength can be borrowed across the cross-section and across related line items. To our knowledge, supervised low-rank tensor regression has not previously been applied to forecasting the full joint panel of firm fundamentals.

Our design philosophy throughout this and the following sections is deliberately conservative. Model architectures and hyperparameters are selected once, on a small development universe, and then locked; all subsequent evaluations—a calendar-fixed holdout, a transfer test to the full cross-section, and a sequence of economic hypotheses specified *ex ante*—reuse the frozen configurations without further tuning. The declarations are timestamped in the version-controlled research log rather than in a formal registry. Where a hypothesis fails, we report the failure.

The empirical work uses three deliberately distinct data vintages. The MFI/FCIX comparison uses fundamentals from 1990Q1–2024Q4 because the FCIX benchmark ends in 2024Q4; the prediction universe is selected using a fixed 2024Q4 market-cap snapshot; and the predictive and economic holdouts extend the fundamentals panel through 2026Q2, with the calendar-fixed training boundary unchanged.

Table 4: Research-design timeline. Dates refer to timestamped entries in the version-controlled research log.

Stage	Frozen before evaluation	Role in the paper
Prediction architecture	v3 residual and ridge-booster objectives locked after the June 20 holdout audit	Confirmatory holdout and transfer evidence
Locked hyperparameters	Rank-1 cells from the June 20/29 summaries, reused without re-tuning	Input to all downstream analyses
499-firm transfer gate	Declared July 6 before the wide-universe refit	Falsification of a mega-cap-only explanation
Veer credit hypothesis	Declared July 6 before the 499-firm veer run	Confirmatory H1/H2 tests
High-yield CDS test	Declared July 7 before HY estimation	Confirmatory traded-credit extension
Secondary routing tests	Added only after headline outcomes were fixed	Exploratory/subordinate robustness

4.1 Tensor construction and imputation

From the 40 fundamental features identified in Section 2.1, we construct a quarterly predictive panel using the cleaned Compustat extract. Year-to-date cash-flow fields are converted to quarterly flows by within-fiscal-year differencing, while genuinely quarterly and trailing-twelve-month fields are retained as reported. The development universe on which all hyperparameters are tuned consists of the top $N = 50$ firms by Compustat market value (`mkvalttq`) at 2024Q4, a reproducible large-cap set that

keeps the search computationally tractable; Section 5 extends the locked models to the full cross-section selected from the top 499 firms. We observe the development firms over 2005Q1–2024Q4, snapping report dates to a strict quarterly grid and keeping the last report per firm-quarter. This produces a raw tensor with shape $50 \times 40 \times 80$ and observed-entry density approximately 90.3%. Each feature is transformed by the log-modulus map

$$\phi(x) = \text{sgn}(x) \log(1 + |x|), \quad (22)$$

which stabilizes scale while preserving sign information. The resulting raw tensor is $\mathcal{X} \in \mathbb{R}^{N \times F \times T}$.

For each time index t , we form an input window and target:

$$\mathcal{X}_t = \mathcal{X}_{:, :, t:t+L-1} \in \mathbb{R}^{N \times F \times L}, \quad \mathbf{Y}_t = \mathcal{X}_{:, :, t:t+L} \in \mathbb{R}^{N \times F}, \quad (23)$$

with observation masks $\mathcal{M}_t$ and $\mathbf{M}_t$ for the input and target, respectively. Stacking across all valid time indices yields a four-dimensional input tensor $\mathcal{X}_{\text{all}} \in \mathbb{R}^{T' \times N \times F \times L}$ and a three-dimensional target $\mathbf{Y}_{\text{all}} \in \mathbb{R}^{T' \times N \times F}$.

Missing entries inside each input window are imputed with a mask-aware Tucker decomposition,

$$\min_{\mathcal{H}_t, \mathbf{A}_t, \mathbf{B}_t, \mathbf{C}_t} \|\mathcal{M}_t \odot (\mathcal{X}_t - \mathcal{H}_t \times_1 \mathbf{A}_t \times_2 \mathbf{B}_t \times_3 \mathbf{C}_t)\|_F^2, \quad (24)$$

where $\mathcal{M}_t$ restricts the objective to observed entries. The fitted Tucker tensor is used only to fill missing entries; observed accounting values are preserved exactly in the covariate tensor passed to the regression step. This distinction is important because the imputer is intended to complete sparse fundamentals, not to smooth or denoise reported values.

The Tucker ranks are selected by a held-out-entry validation procedure. For each rolling window, we hide 10% of observed entries, stratified by feature, fit the mask-aware Tucker model on the remaining observed entries, and score relative Frobenius error only on the hidden entries. The time-mode rank is fixed at L , and the firm- and feature-mode ranks are swept over a small grid. This procedure selects ranks $(2, 2, 2)$ for $L = 2$ and $(4, 4, 4)$ for $L = 4$; the $L = 2$ choice is the parsimonious one-standard-error selection, while $(4, 4, 4)$ minimizes the validation error for the one-year lookback. The final imputation step uses SVD initialization, a maximum of 100 iterations, and tolerance 10^{-5} . Figure 5 illustrates the decomposition structure. We compute an observed-only RMS scale for each window,

$$r_t = \sqrt{\frac{1}{|\Omega_t|} \sum_{(i,j,k) \in \Omega_t} x_{ijk}^2}, \quad (25)$$

where Ω_t is the set of observed entries in $\mathcal{X}_t$. We evaluate two input-scaling variants: an unscaled covariate tensor in which missing entries are imputed in original units and observed entries remain unchanged (denoted Unscaled), and an RMS-normalized covariate tensor in which the completed input window is divided by r_t (denoted Normalized).

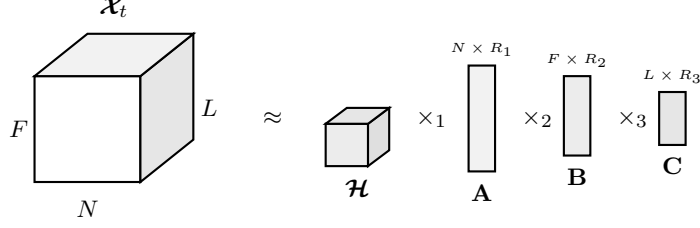


Fig. 5: Mask-aware Tucker imputation for sparse fundamentals. To resolve missing entries without discarding sparse features, the incomplete input tensor $\mathcal{X}_t \in \mathbb{R}^{N \times F \times L}$ is approximated by a dense core tensor $\mathcal{H} \in \mathbb{R}^{R_1 \times R_2 \times L}$ contracted with orthogonal factor matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ along each respective mode. The decomposition is fitted on observed entries only; observed values are preserved exactly, and only missing cells are filled from the low-rank reconstruction. The validated ranks are $(R_1, R_2, L) = (2, 2, 2)$ for the two-quarter lookback and $(4, 4, 4)$ for the four-quarter lookback.

4.2 Low-rank CP regression as a booster

The estimator is organized as a baseline-plus-booster stack: a simple, well-understood baseline predictor absorbs the level and persistence of each firm–feature series, and a low-rank CP tensor regression is fitted to the baseline’s forecast residuals, capturing whatever cross-firm and cross-feature structure the baseline cannot represent. The final prediction is

$$\hat{\mathbf{Y}}_t = \hat{\mathbf{B}}_t + \gamma \cdot s \cdot \langle \mathcal{X}_t, \mathcal{W} \rangle \in \mathbb{R}^{N \times F}, \quad (26)$$

where $\hat{\mathbf{B}}_t$ is the baseline forecast, $\gamma \geq 0$ is a tuned mixing weight, and s inverts the scaling applied to the residual target during training. This architecture answers the question that matters for our purposes: does the multilinear structure add predictive information beyond what a strong conventional model already delivers? We instantiate the stack with two baselines of increasing strength.

Fixed-effects cell. The first baseline is a within-firm centering analogous to fixed effects in panel econometrics. For each firm–feature pair, we compute a mask-aware temporal mean over the training windows,

$$\mu_{ij} = \frac{\sum_t m_{tij} y_{tij}}{\sum_t m_{tij} + \varepsilon}, \quad (27)$$

where m_{tij} is the observation indicator and $\varepsilon = 10^{-8}$ prevents division by zero, and set $\hat{\mathbf{B}}_t = \boldsymbol{\mu}$. The CP target is the centered residual, normalized by its observed root-mean-square

$$s = \sqrt{\frac{1}{|\Omega|} \sum_{(t,i,j) \in \Omega} (y_{tij} - \mu_{ij})^2}, \quad (28)$$

where $\Omega = \{(t, i, j) : m_{tij} = 1\}$, yielding $\tilde{y}_{tij} = m_{tij}(y_{tij} - \mu_{ij})/s$; missing entries are set to zero in the scaled target. Optional per-feature target and input rescalings are exposed as tuned binary switches.

Ridge-booster cell. The second baseline is the full per-feature ridge regression described in Section 4.3—the same model that serves as our competitive benchmark. Here the CP tensor is trained on the ridge model’s out-of-fold residuals: within each training set, an inner three-fold time-series split produces ridge predictions for every window from models that never saw that window, and the CP target is $\tilde{y}_{tij} = m_{tij}(y_{tij} - \hat{y}_{tij}^{\text{ridge, OOF}})/s$. Windows without valid out-of-fold coverage are excluded from CP training rather than back-filled. This out-of-fold discipline prevents the booster from learning the baseline’s in-sample optimism, and at prediction time the baseline is refitted on the full training set.

In both cells, the residual map is a low-rank CANDECOMP/PARAFAC (CP) tensor regression. The coefficient tensor $\mathcal{W} \in \mathbb{R}^{N \times F \times L \times N \times F}$ admits the CP form

$$\mathcal{W} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r \circ \mathbf{d}_r \circ \mathbf{e}_r, \quad (29)$$

where $\mathbf{a}_r \in \mathbb{R}^N$, $\mathbf{b}_r \in \mathbb{R}^F$, and $\mathbf{c}_r \in \mathbb{R}^L$ correspond to the input modes, while $\mathbf{d}_r \in \mathbb{R}^N$ and $\mathbf{e}_r \in \mathbb{R}^F$ correspond to the output modes of the r -th rank-one component. Figure 6 illustrates the structure of the CP regression model.

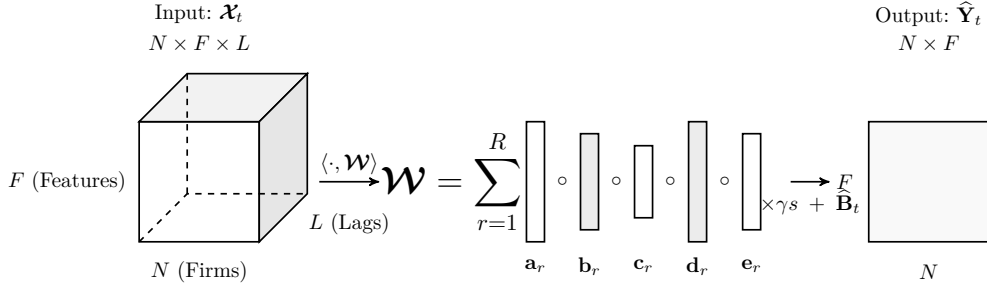


Fig. 6: Low-rank CP tensor regression model. The 5D weight tensor $\mathcal{W}$ is parameterized as a sum of R rank-one components. The multilinear contraction $\langle \mathcal{X}_t, \mathcal{W} \rangle$ maps the historical 3D input ($N \times F \times L$) to a 2D next-quarter residual correction ($N \times F$), which is rescaled by γs and added to the baseline forecast $\hat{\mathbf{B}}_t$ (firm–feature fixed effects or per-feature ridge).

We estimate $\mathcal{W}$ via alternating least squares in TensorLy [9] by regularized least squares,

$$\min_{\mathcal{W}} \sum_t \left\| \tilde{\mathbf{Y}}_t - \langle \mathcal{X}_t, \mathcal{W} \rangle \right\|_F^2 + \lambda \|\mathcal{W}\|_F^2, \quad (30)$$

where $\tilde{\mathbf{Y}}_t$ is the zero-filled, centered, RMS-scaled residual target defined above. Because the CP solver does not accept observation masks, missing entries are set to zero prior to fitting; all evaluation at test time remains fully mask-aware.

Hyperparameters are tuned with Optuna [10] using a tree-structured Parzen estimator over the CP rank $R \in \{1, \dots, 25\}$, the ridge penalty $\lambda \in [10^{-2}, 10^5]$ (log-uniform), the mixing weight $\gamma \in [0, 2]$, and three binary scaling switches (global RMS scaling of the residual target, per-feature target scaling, and per-feature input scaling). The search objective is the incremental fit

$$\Delta R^2 = R^2(\mathbf{Y}, \widehat{\mathbf{B}} + \gamma \widehat{\mathbf{CP}}) - R^2(\mathbf{Y}, \widehat{\mathbf{B}}), \quad (31)$$

averaged over a three-fold time-series split of the development block (the first 80% of rolling windows), where R^2 is the pooled mask-aware statistic defined in Section 4.3. Scoring the increment over the cell’s own baseline—rather than raw fit—forces the search toward genuinely complementary tensor structure. Each of the four cells (two baselines $\times L \in \{2, 4\}$) received several thousand trials distributed across a compute cluster; the top-five trials per cell form a tight plateau, indicating that the search converged. The rank-1 trial per cell is then locked and used unchanged in every subsequent analysis in this paper.

4.3 Ridge benchmark and evaluation protocol

The benchmark against which the tensor must prove itself is a per-feature ridge regression on vectorized lag windows; the same model also serves as the baseline inside the ridge-booster cell of Section 4.2. Both the CP and ridge models share the same within-firm centering and the same zero-fill convention for missing target cells; the key distinction is that each ridge model is fitted independently for a single feature with its own regularization penalty, whereas the CP tensor shares one low-rank parameterization across all features simultaneously. For each feature $j \in \{1, \dots, F\}$, let $\mathbf{y}^{(j)} \in \mathbb{R}^{T'N}$ denote the stacked target vector and $\mathbf{X} \in \mathbb{R}^{T'N \times FL}$ the design matrix formed by flattening the lag dimension. The per-feature model is

$$\mathbf{y}^{(j)} = \mathbf{X}\boldsymbol{\beta}^{(j)} + \boldsymbol{\mu}^{(j)} + \boldsymbol{\epsilon}^{(j)}, \quad (32)$$

where $\boldsymbol{\mu}^{(j)}$ contains the firm fixed effects for feature j .

Our pipeline computes mask-aware firm-feature means $\mu_i^{(j)}$ and models the de-meaned target directly—the classical within-transformation in panel econometrics—removing the fixed effects before estimation and restoring them at prediction time.

To make the panel structure explicit, let $\mathbf{y}_i^{(j)} \in \mathbb{R}^{T'}$ denote the time-series target for firm i and feature j , and let $\mathbf{X}_i \in \mathbb{R}^{T' \times FL}$ denote the firm’s design matrix. The stacked panel regression groups observations by firm:

$$\begin{bmatrix} \mathbf{y}_1^{(j)} \\ \mathbf{y}_2^{(j)} \\ \vdots \\ \mathbf{y}_N^{(j)} \end{bmatrix} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_N \end{bmatrix} \boldsymbol{\beta}^{(j)} + \begin{bmatrix} \boldsymbol{\mu}_1^{(j)} \\ \boldsymbol{\mu}_2^{(j)} \\ \vdots \\ \boldsymbol{\mu}_N^{(j)} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\epsilon}_1^{(j)} \\ \boldsymbol{\epsilon}_2^{(j)} \\ \vdots \\ \boldsymbol{\epsilon}_N^{(j)} \end{bmatrix}, \quad (33)$$

where $\boldsymbol{\mu}_i^{(j)} \in \mathbb{R}^{T'}$ replicates firm i 's historical mean across every time step. For a generic firm i , the internal elements of $\mathbf{y}_i^{(j)}$ and $\mathbf{X}_i$ expand across time steps, features, and lags as:

$$\mathbf{y}_i^{(j)} = \begin{bmatrix} y_{i,t+1}^{(j)} \\ \vdots \\ y_{i,T'}^{(j)} \end{bmatrix}, \quad \mathbf{X}_i = \begin{bmatrix} \overbrace{x_{i,t}^{(1)} \cdots x_{i,t}^{(F)}}^{\text{Lag } 1} & \cdots & \overbrace{x_{i,t-L+1}^{(1)} \cdots x_{i,t-L+1}^{(F)}}^{\text{Lag } L} \\ \vdots & \ddots & \vdots \\ x_{i,T'-1}^{(1)} \cdots x_{i,T'-1}^{(F)} & \cdots & x_{i,T'-L}^{(1)} \cdots x_{i,T'-L}^{(F)} \end{bmatrix}. \quad (34)$$

To ensure a rigorously competitive benchmark, ridge penalties are independently optimized for each feature via an inner three-fold time-series cross-validation from the candidate set $\alpha \in \{10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3, 10^4\}$. This guarantees that the benchmark is optimally tuned for the specific scale and variance of every individual fundamental feature. When a feature has too few observations (below an adaptive threshold), we revert to predicting the firm-mean baseline.

Performance is evaluated using a pooled, mask-aware coefficient of determination,

$$R^2 = 1 - \frac{\sum_{t,i,j} m_{tij} (y_{tij} - \hat{y}_{tij})^2}{\sum_{t,i,j} m_{tij} (y_{tij} - \bar{y})^2}, \quad (35)$$

with $\bar{y}$ the mean of observed entries; per-feature R^2 values identify which fundamentals are most predictable.

Calendar-fixed holdout. Final evaluation uses a strict temporal split whose boundary is fixed in calendar time rather than as a fraction of the sample: every window whose prediction target falls in 2021Q1 or later belongs to the test block, and no model, hyperparameter, or scaling constant sees any test-block data. Fixing the boundary by calendar date means that extending the panel forward in time only appends new test quarters; the training set is byte-identical across data refreshes. On the extended panel (fundamentals through 2026Q2), this yields 62 ($L = 2$) or 60 ($L = 4$) training windows and 21 test windows spanning 2021Q1–2026Q2—a genuine out-of-time extrapolation covering the post-pandemic tightening cycle.

4.4 Empirical results

Table 5 reports pooled, mask-aware test R^2 for the four locked cells on the calendar-fixed holdout. In every cell, adding the CP booster improves on its own baseline: by +0.043 to +0.048 over firm–feature fixed effects, and—the more demanding comparison—by +0.011 to +0.019 over the fully tuned per-feature ridge. The latter increment is earned against a benchmark that enjoys a separate, independently regularized model for every feature, while the tensor shares a single low-rank parameterization and one global hyperparameter set across all forty. The result is robust to hyperparameter choice: across the top five Optuna trials in each of the four cells, all twenty holdout deltas are positive, and within the $L = 2$ cells the top-five spread is below 0.002 in R^2 .

Table 5: Locked-cell holdout results: pooled mask-aware R^2 on the calendar-fixed test block (21 windows, 2021Q1–2026Q2; development universe of 50 firms). Each row is the rank-1 Optuna trial of its cell; the CP booster is added to the cell’s own baseline.

Baseline	L	Baseline R^2	Baseline+CP R^2	Δ
Fixed effects	2	0.721	0.768	+0.048
Fixed effects	4	0.722	0.765	+0.043
Ridge	2	0.764	0.783	+0.019
Ridge	4	0.767	0.777	+0.011

Where does the gain come from? A per-feature decomposition of the booster delta, repeated for all twenty top-five trials, shows that the improvement concentrates sharply on non-stationary features. Using two train-set stationarity metrics per feature—the variance ratio $\text{std}(\Delta x_f)/\text{std}(x_f)$ of the cross-sectional mean series (low values indicate trending behavior) and a normalized trend slope—the correlation between a feature’s stationarity and its CP delta is -0.45 to -0.48 (variance ratio) and $+0.44$ to $+0.50$ (trend slope) in every cell, and the non-stationary tercile of features gains six to twelve times more than the stationary tercile. The features that benefit most are large, trending balance-sheet and income items (total long-term debt, net sales, operating income before depreciation, receivables), while features that are already near-perfectly predicted by their own history (inventories, netted liabilities) gain little. The tensor’s value is therefore not diffuse: it captures shared low-rank co-movement precisely where per-feature linear-in-history models are mis-specified.

Two further regularities sharpen the picture. First, the booster delta is largest in test windows with low cross-sectional dispersion of targets (correlation ≈ -0.8 in the ridge cells)—exactly the windows in which firms co-move and a common latent factor exists for the tensor to exploit. Second, the delta does not decay as the forecast origin moves further from the training boundary; if anything it grows slightly across the 21-quarter test block, indicating the structure being captured is persistent rather than an artifact of the split point.

5 Transfer to the Full Cross-Section

A predictive gain established on fifty mega-caps could in principle be an artifact of that particularly well-measured corner of the market. We therefore refit all four locked cells—hyperparameters frozen, no re-tuning of any kind—on the full cross-section selected from the top 499 firms by 2024Q4 market value, using fundamentals through 2026Q2 and the same calendar-fixed split. One selected firm has insufficient cache coverage, so the realized transfer tensor contains 498 firms. The ex ante criterion was directional: if the CP increment collapsed to zero or below in all four cells, that collapse would itself be the finding.

Table 6 shows the increment survives in all four cells. Against the ridge benchmark, the transfer delta is essentially unchanged from the development universe; against fixed effects it remains clearly positive but shrinks to roughly 40% of its mega-cap

Table 6: Universe transfer at locked hyperparameters: pooled mask-aware test R^2 for the 498 firms that survive cache construction from the 499-firm selection universe (22 test windows), against the 50-firm development deltas.

Baseline	L	Baseline R^2	+CP R^2	Δ (499)	Δ (50)
Ridge	2	0.745	0.759	+0.014	+0.019
Ridge	4	0.751	0.762	+0.011	+0.011
Fixed effects	2	0.691	0.709	+0.018	+0.048
Fixed effects	4	0.693	0.713	+0.020	+0.043

size. The multilinear structure is therefore not a mega-cap artifact, though its edge over weak baselines is strongest where fundamentals are best measured—a pattern that recurs throughout the economic analysis below.

Computationally, the tenfold increase in the firm mode makes the stock CP-ALS implementation infeasible: materializing the Khatri–Rao design matrix at 498 firms exhausts memory. We instead exploit the fact that the ALS normal equations factorize exactly—the Gram matrix of the design factorizes into a Hadamard product of small per-mode Gram matrices—so no large intermediate is ever formed. The resulting solver is algebraically identical to the reference implementation (same initialization, same ridge solves, same convergence rule), reduces peak memory by two orders of magnitude, and was verified to reproduce the stock solver’s output on the development universe.

6 Economic Content of Tensor Forecasts

An out-of-sample R^2 improvement in accounting space says nothing, by itself, about economic significance. This section asks whether the tensor forecasts carry information that markets price, and if so, where. Our protocol is uniform: for each hypothesis we declare the signal, target, controls, significance criterion, and interpretation before running the test, and we report every declared outcome, including the null ones. All tests are run separately in each of the four locked cells; agreement across cells (which share data but differ in architecture) is treated as robustness, not as independent replication.

6.1 Announcement event studies: returns, volatility, and option prices

The first battery aligns test-set forecasts with quarterly earnings announcements on the development universe (960 scored announcement events per cell over 21 test quarters). The ex ante signal is the magnitude of the tensor’s forecast adjustment: the standardized absolute difference between the ensemble and baseline predictions, available before the announcement. Outcomes include announcement-window abnormal returns and post-announcement realized and idiosyncratic volatility over the following 30 trading days, tested per feature–target pair with cross-sectional regressions and a Benjamini–Yekutieli false-discovery-rate control [11] at $q < 0.1$ over the full pair grid.

Three findings are unambiguous. First, abnormal announcement returns are a clean null: zero of the ex ante return-family pairs survive FDR control, in all four cells. Whatever the tensor knows about fundamentals, it does not predict announcement-window price reactions in mega-caps. Second, the forecast-adjustment magnitude does forecast post-announcement volatility beyond lagged realized volatility, but this is architecture-dependent: controlling for pre-event volatility, the signal remains significant ($|t| \geq 2$) in 75 and 69 of 80 pairs in the two ridge-booster cells, against only 13 and 6 in the fixed-effects cells. Third, and decisively for tradeability, the volatility information is subsumed by option prices: adding the pre-event at-the-money implied volatility from OptionMetrics as a control eliminates the majority of significant pairs in every cell (at best 29 of 80 survive, in most cells far fewer), and a long-short straddle proxy sorted on the signal against the implied benchmark earns uniformly negative or mixed means—the mechanical variance risk premium, not alpha. In short, the tensor’s volatility signal is real but already embedded in option prices for these firms.

6.2 Forecast-error signals and credit: the veer framework

The event studies use the forecast’s magnitude; the richer object is the forecast error—the direction in which a firm’s realized fundamentals veer away from the model’s expectation. For each firm, feature, and test quarter we form a studentized forecast surprise: the realized-minus-predicted error, de-biased by an expanding firm-level median of past errors and scaled by an expanding per-feature median absolute deviation (using only past test quarters, with a four-quarter burn-in). Feature-level surprises are aggregated into five themed scores (leverage, earnings, investment, balance-sheet liquidity, and cash flow) following the natural accounting groupings of the 40 features, and a three-quarter rolling mean of each themed score defines a drift signal capturing persistent, directional veering. All signals are ex ante: they use only information available after quarter t ’s announcement to predict outcomes over the following quarter.

On the 499-firm selection universe (roughly 6,900 scored events per cell over 18 test quarters), we specified one headline hypothesis before evaluation: persistent cash-flow over-performance against the tensor forecast predicts improving credit quality, measured as the change in a naive Merton distance-to-default [12] from two trading days before the announcement to 63 trading days after, controlling for the pre-event level and pre-trend. The test is a Fama–MacBeth regression [13] across announcement quarters, with a partial rank information coefficient as a secondary check, and a declared success criterion of a positive slope at one-sided $p < 0.05$ in at least two of four cells.

The hypothesis was confirmed; Table 7 reports the four cells. The Fama–MacBeth slope is +0.020 in all four cells, with one-sided $p = 0.033$ and 0.040 in the two fixed-effects cells, $p = 0.060$ and 0.071 in the ridge cells, and a positive partial rank IC ($t \approx +2.1$ to +2.6) throughout.

In economic units, a one-standard-deviation cash-flow drift maps to a distance-to-default improvement of +0.05 to +0.07; evaluated on the panel’s distance-to-default distribution, this is a 29% to 56% relative reduction in the model-implied physical

Table 7: Credit test specified ex ante on the 499-firm selection universe: Fama–MacBeth slope of the post-announcement change in distance-to-default on the cash-flow drift signal, with pre-event level and pre-trend controls ($\approx 6,900$ events per cell; declared sign positive, one-sided p ; success bar: $p < 0.05$ in ≥ 2 of 4 cells).

Baseline	L	Slope	t	One-sided p	Partial IC t
Ridge	2	+0.0204	+2.06	0.060	+2.1
Ridge	4	+0.0197	+1.97	0.071	+2.2
Fixed effects	2	+0.0202	+2.28	0.040	+2.4
Fixed effects	4	+0.0204	+2.39	0.033	+2.6

default probability, growing with the firm’s initial credit quality. For this investment-grade universe, however, absolute default probabilities are tiny, so a large relative change need not be reflected in traded prices; the next subsection tests exactly that. The secondary hypothesis specified ex ante (that the joint set of veer signals adds out-of-sample explanatory power for post-announcement changes in implied volatility) passed formally in three of four cells but with increments of at most +0.003 in R^2 ; we treat it as economically negligible, consistent with the option-subsumption result above.

6.3 Traded credit: an investment-grade null and a high-yield confirmation

To test whether the credit information is priced, we extended the confirmed hypothesis to market credit-default-swap spreads: five-year single-name CDS (senior unsecured, XR/XR14 documentation) from Markit, matched to 169 of the 499 firms with a name-validated link. The declared target was the change in log spread over the same event window, with the same controls; expected sign negative (over-performance tightens spreads). The result is a clean null in all four cells: slopes are indistinguishable from zero ($|t| < 0.15$), amounting to ± 0.03 basis points per one-standard-deviation signal at the sample’s 56 basis-point median spread. The null is not a matching artifact: re-running the distance-to-default regression on exactly the CDS-covered subsample leaves the slope intact and slightly larger (+0.024 to +0.026, $t > 2$). The same firm-events whose model-implied credit quality improves show no movement in their traded spread. For investment-grade mega-caps, every traded price we test—equity around announcements, options on volatility, and CDS on credit—already embeds what the tensor forecast errors know; only model-implied quantities respond. Within the investment-grade panel, however, the distance-to-default effect concentrates in the highest spread tercile in all four cells, suggesting the channel matters where default risk is material.

We therefore constructed a dedicated high-yield and crossover universe, again declaring the hypothesis before estimation. From the U.S. five-year senior CDS reference entities in Markit, we selected issuers with median spreads of at least 150 basis points and matched them to listed CRSP/Compustat firms, yielding a universe of 113

names with a median spread of 289 basis points, against 56 in the investment-grade sample.² Because the universe is defined at 2024Q4 it conditions on survival, which excludes defaulted names and biases against the hypothesis. The full pipeline was re-estimated at locked hyperparameters; the CP increment remains positive in three of four cells, turning negative in the fixed-effects $L = 4$ cell, consistent with low-rank structure weakening as fundamentals become noisier down the credit spectrum.

Table 8: High-yield CDS test specified ex ante: Fama–MacBeth slope of the post-announcement change in log five-year CDS spread on the cash-flow drift signal, with pre-event spread level and pre-trend controls (900 events, 90 CDS-covered firms, 13 quarters per cell; declared sign negative, one-sided p). Basis-point impact evaluated at the 282 bp sample median spread.

Baseline	L	Slope	t	One-sided p	bp per 1σ
Ridge	2	−0.0113	−2.21	0.024	−6.3
Ridge	4	−0.0151	−2.74	0.009	−6.8
Fixed effects	2	−0.0112	−2.34	0.019	−6.9
Fixed effects	4	−0.0104	−1.96	0.037	−6.4

We subject this CDS result to a small robustness battery, summarized in Table 9. The sign is stable: the cash-flow drift coefficient is negative in every cell under log-spread changes at 21-, 42-, and 63-trading-day horizons, under raw-basis-point spread changes, after one-quarter-at-a-time deletions, and within each pre-event spread tercile. Statistical strength is not uniform. The 42-day window is strongest; raw spread changes preserve the economic sign and imply larger basis-point moves but are noisier; 5% winsorization and dropping 2022 tightening quarters attenuate significance in some cells. The tercile split is economically useful: the basis-point effect is largest in the widest-spread tercile, consistent with the signal mattering most where default risk is closest to the money.

Table 8 reports the outcome specified ex ante: the hypothesis is confirmed in all four cells against a declared bar of two. In crossover and high-yield names, a one-standard-deviation persistent cash-flow over-performance against the tensor forecast predicts approximately 2.3% of spread tightening—about 6.5 basis points at the median spread—over the following quarter, controlling for the spread level and its pre-trend, with negative partial rank ICs in all four cells. The same signal that moved only a model-implied credit metric in the investment-grade universe moves the traded market price of credit here. Together, the two CDS experiments are consistent with faster incorporation of this information in mega-cap investment-grade markets and slower incorporation where default risk is material. Given the small sample (90 firms,

²Matching is by normalized issuer name, with financing subsidiaries mapped to their listed parents and private or defunct entities excluded. Of 970 reference entities, 298 met the spread threshold and 128 matched, of which 113 had sufficient fundamentals coverage.

Table 9: High-yield CDS robustness summary. Entries aggregate the four locked cells. The sign column reports the number of cells with the expected negative slope; the basis-point range is the implied effect of a one-standard-deviation cash-flow drift at the relevant median spread.

Variant	Expected sign	bp range	Comment
Log spread, +21td	4/4	−3.9 to −4.4	shorter-window effect
Log spread, +42td	4/4	−6.3 to −6.7	strongest t-statistics
Log spread, +63td	4/4	−6.3 to −6.9	headline window
Raw spread change, +63td	4/4	−12.4 to −14.5	noisier in levels
1% winsorized, +63td	4/4	−5.6 to −6.4	sign intact
5% winsorized, +63td	4/4	−4.0 to −5.1	attenuated
Drop 2022 quarters	4/4	−5.4 to −6.5	reduced power
Leave-one-quarter-out	4/4	−5.0 to −8.6	sign never flips
Widest spread tercile	4/4	−11.9 to −18.4	largest economic effect

13 quarters) and the correlation across cells, we count this as a single confirmation rather than four independent ones.

Three secondary results complete the picture. First, an options-tradeability battery was specified ex ante on the high-yield universe, using the same straddle-proxy long–short construction that failed at mega-caps: the primary criterion—a positive and bootstrap-significant long–short straddle mean in at least two of four cells—was met exactly at the declared bar, but only in the two $L = 4$ cells, and the secondary criterion (per-pair significance incremental to implied volatility) failed in all four. We interpret this as a weak, configuration-dependent effect and do not claim a second tradeable result. Second, the drift signal does not predict forward equity returns in the 499-firm selection universe (Fama–MacBeth $t < 0.8$ everywhere; tercile long–short spreads positive but insignificant), closing the equity route in this universe for both the magnitude and error-direction signals. Third, forecast errors are essentially idiosyncratic: clustering firms by error co-movement recovers no sectoral structure (adjusted Rand index versus GICS below 0.04 in every cell) and the first principal component explains under 4% of error variance—the veer signals are firm-specific news, not a missed common factor.

7 Conclusion

This paper develops a tensor-based framework for firm fundamentals in three movements. First, treating the panel of accounting data as a third-order tensor, we constructed the Market Fundamentals Index from the temporal factors of a mask-aware Tucker decomposition and showed that this fundamentals-based volatility measure is statistically dependent on price-based chaos (FCIX), with independence rejected at the 1% level.

Second, we showed the same tensor structure is predictive. A low-rank CP tensor regression, deployed as a booster over per-feature ridge and fixed-effects baselines with strict out-of-fold discipline, improves out-of-time forecasts of next-quarter fundamentals in every locked configuration, on both a 50-firm development universe and the wider 499-firm selection universe at frozen hyperparameters. The gain concentrates

precisely on non-stationary, trending line items and on periods of cross-sectional co-movement—the regimes where per-feature linear models are mis-specified and shared low-rank structure exists to be exploited.

Third, we asked whether markets price what the tensor knows, using hypotheses, criteria, and interpretations specified before evaluation and timestamped in the research log. In mega-cap investment-grade names, every traded price we examined—announcement returns, option-implied volatility, and CDS spreads—already embeds the information in the tensor’s forecasts and forecast errors; only model-implied quantities (a Merton distance-to-default) respond, and a large relative improvement in an infinitesimal default probability is worth little at the spread. In crossover and high-yield credit, where default risk is material, the ex ante CDS test confirmed in all four model configurations that persistent cash-flow over-performance against the tensor forecast predicts economically meaningful spread tightening—roughly 6.5 basis points per standard deviation per quarter at the median spread—controlling for spread level and momentum. The pattern is consistent with slower incorporation of this information lower down the credit spectrum.

The analysis has candid limitations that chart the path forward: the high-yield sample is short and survivorship-gated, the four model cells are correlated rather than independent replications, and the equity and options routes remain closed or fragile. Natural extensions include longer high-yield histories, dead-firm inclusion to remove the survivorship gate, and routing the veer signals to credit instruments directly tradeable at scale.

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